

Horizontal Gevrey Quasianalyticity Along Hörmander Orbits

Candidate partial answer to Question 3 of arXiv:1805.08667

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Status. Candidate substantial partial result, likely valid, subject to human review. It settles the two horizontal-derivative definitions generally and all three definitions on the Heisenberg group. The general sub-Laplacian-defined class remains outside the proved scope.

1 Source question and scope

Fischer–Ruzhansky–Taranto introduce three subelliptic Gevrey classes from a globally defined Hörmander family $\mathbf{X} = \{X_1, \dots, X_r\}$: bounds on all horizontal words in L^∞ , the analogous bounds in L^2 , and bounds on iterates of the sub-Laplacian $\mathcal{L} = -\sum_j X_j^2$ [1]. Their Question 3 asks how these spaces relate to elliptic Gevrey spaces and, at order $s = 1$, whether they too can contain no nonzero compactly supported functions.

may still broadly use our methods as explicit formulae close to the ones we use are known [8, 2], we suspect that the investigations on stratified nilpotent Lie groups of any step will require new tools.

Focussing on the case of stratified nilpotent Lie groups at first is natural as it gives a rich setting where the Hörmander family $\mathbf{X}$ and the associated family are well understood. Furthermore, it is possible that the Rothschild–Stein machinery - also known as nilpotentisation - may help transfer results from the nilpotent case to the sub-Riemannian manifold setting.

3. Is it possible to relate sub-elliptic Gevrey spaces with other spaces? For instance, can they be related to elliptic Gevrey spaces? In the case $s = 1$, the Euclidean Gevrey spaces are exactly the space of analytical functions and therefore do not contain any compactly supported functions. Can it be the same for sub-elliptic Gevrey spaces?

We end this introduction with a word on our motivations to define these new functional spaces and on their applications. Beside the ones already mentioned for wave equations above, a long-term goal would be to study the Gevrey hypoellipticity of a sum of square of vector fields on an open set of the Euclidean space or of sub-Laplacians on sub-Riemannian manifolds. This is still an open question with many contributions for which we will cite [3, 31] and refer to all the references therein; note that Gevrey hypoellipticity and solvability of linear partial differential operators in the Euclidean context or on Gevrey manifolds is quite well understood [1]. It is not difficult to see that a sub-Laplacian $\mathcal{L}$ will be hypoelliptic for the subelliptic Gevrey spaces

Figure 1: Question 3 in the source paper (PDF p. 4).

The theorem below completely answers the compact-support clause for the two horizontal definitions. The source’s own equivalence theorem then answers it for the sub-Laplacian definition on $\mathbb{H}_n$. We isolate the still-open general spectral scope in Section 4.

2 Horizontal quasianalyticity

For a word $I = (i_1, \dots, i_m)$, write $X_I = X_{i_1} \cdots X_{i_m}$. Following the source, if the index j occurs α_j times, put $I! = \prod_j \alpha_j!$. Thus the repeated word $(j, \dots, j)$ has $I! = m!$.

Proof intuition

The full noncommutative word structure is not needed for uniqueness. Repeating just one generator shows that the restriction of the function to each X_j -integral curve has ordinary analytic derivative bounds. An open zero set therefore travels intact along every local flow. Finite compositions of those flows generate the Hörmander orbit, and bracket generation makes that orbit all of a connected manifold. The argument uses neither analytic coefficients nor estimates for higher-order Riesz transforms.

Theorem 1 (Open-set uniqueness). *Let M be a connected smooth manifold and $\mathbf{X} = \{X_1, \dots, X_r\}$ a finite family of smooth real vector fields whose Lie brackets span $T_p M$ at every $p \in M$. Suppose $\phi \in C^\infty(M)$ and, for some $C, A > 0$ and $0 < s \leq 1$,*

$$\|X_I \phi\|_{L^\infty(M)} \leq C A^{|I|} (I!)^s \quad \text{for every word } I. \quad (1)$$

If ϕ vanishes on a nonempty open subset of M , then $\phi \equiv 0$.

Proof. Fix j , a point p , and a maximal integral curve $\gamma(t) = \text{Fl}_{X_j}^t(p)$, defined on an interval J . For $u(t) = \phi(\gamma(t))$, the chain rule gives

$$u^{(m)}(t) = X_j^m \phi(\gamma(t)).$$

The repeated word has factorial $m!$, so (1) and $s \leq 1$ give

$$|u^{(m)}(t)| \leq C A^m (m!)^s \leq C A^m m!.$$

Taylor's formula therefore bounds the remainder after order N , whenever the segment from t to $t+h$ lies in J , by

$$|R_N(t, h)| \leq \frac{C A^{N+1} (N+1)!}{(N+1)!} |h|^{N+1} = C(A|h|)^{N+1}.$$

Thus u is real analytic on J . In particular, if it vanishes on any nonempty subinterval, it vanishes on all of J .

Let U be a nonempty open set on which $\phi = 0$. The preceding paragraph shows that, for every local flow time for which it is defined, $\text{Fl}_{X_j}^t(U)$ is again an open zero set. The same is true after any finite composition of local flows of the X_j 's. Hence the flow saturation of U is contained in $\{\phi = 0\}$.

By the orbit theorem, the orbit of the family $\mathbf{X}$ is an immersed submanifold whose tangent space contains the evaluations of all iterated Lie brackets [2]. The Hörmander hypothesis makes every orbit open. These orbits partition M ; connectedness therefore forces a single orbit. The saturation of U is all of M , so $\phi \equiv 0$. $\square$

Corollary 2 (Proper compact support). *Under the hypotheses of Theorem 1, no nonzero function obeying (1) can have proper compact support. In particular, if M is connected and noncompact, then*

$$\gamma_{\mathbf{X}, L^\infty}^s(M) = \{0\}, \quad 0 < s \leq 1,$$

with the source's convention that this class consists of compactly supported smooth functions.

Proof. The complement of a proper support is a nonempty open zero set. On a noncompact manifold every compact subset is proper. $\square$

3 Transfers to the other source definitions

Corollary 3 (Horizontal L^2 class). *Assume in addition that the horizontal Sobolev embedding of Definition 2.5 in the source holds. Then a compactly supported member of $\gamma_{\mathbf{X},L^2}^s(M)$, $0 < s \leq 1$, which vanishes on a nonempty open set is zero. Consequently, on a connected noncompact M , that class has no nonzero compactly supported member.*

Proof. Proposition 2.6 of the source shows that a compactly supported member of the horizontal L^2 class belongs to the horizontal L^∞ class, with the same Gevrey order [1]. Apply Theorem 1. $\square$

The paper verifies the required Sobolev embedding for Lie groups of polynomial volume growth, so Corollary 3 applies throughout that principal noncompact setting.

Corollary 4 (All three definitions on $\mathbb{H}_n$). *Let $\mathcal{L}$ be the standard sub-Laplacian on the Heisenberg group $\mathbb{H}_n$. For every $0 < s \leq 1$,*

$$C_c^\infty(\mathbb{H}_n) \cap \gamma_{\mathcal{L}}^s(\mathbb{H}_n) = \{0\}.$$

The same conclusion holds for both horizontal classes in the source.

Proof. Theorem 4.1 of the source proves $\gamma_{\mathcal{L}}^s(\mathbb{H}_n) = \gamma_{\mathbf{X},L^2}^s(\mathbb{H}_n)$ for every $s > 0$, and it identifies the two horizontal definitions on compactly supported functions [1]. Corollaries 2 and 3 finish the proof. $\square$

Thus the answer to the compact-support clause of Question 3 is affirmative for the horizontal definitions on every connected noncompact Hörmander manifold in the stated setting, and for all three definitions on the Heisenberg groups.

4 Verification, limitations, and novelty

The proof uses only four checkable ingredients: the repeated-word identity $I! = m!$, differentiation along a flow, Taylor’s remainder, and the orbit theorem. It works for smooth vector fields; analytic coefficients and completeness of the flows are unnecessary. It also strengthens the questioned endpoint from $s = 1$ to the whole range $0 < s \leq 1$.

The remaining limitation is substantive. For a general sub-Laplacian, the source only proves $\gamma_{\mathbf{X},L^2}^s \subseteq \gamma_{\mathcal{L}}^s$. Reversing this inclusion requires word-uniform higher-order Riesz-transform estimates and is essentially Question 2 of the same paper. A bound on $\mathcal{L}^k \phi$ alone therefore cannot be inserted into the flow proof without assuming precisely the missing upgrade. The packet makes no claim for the general $\gamma_{\mathcal{L}}^s$ class beyond settings, such as $\mathbb{H}_n$, where the equivalence is known.

On a compact manifold the phrase “compactly supported” is vacuous: constants are compactly supported and belong to these classes. Open-set uniqueness is the invariant statement, and proper compact support is the correct corollary.

Novelty search. On 17 August 2026, bounded searches covered the run’s registry, solution, attempt, and proof-gap indexes and arXiv queries by exact title, arXiv id, authors, the compact-support question, and close horizontal Gevrey/quasianalytic/Hörmander-flow phrases. No later explicit answer to Question 3 or duplicate of this argument was found. The mechanism is elementary, so novelty confidence is moderate rather than high.

Human-review recommendation. Check the source’s word-factorial convention, the continuation from a local zero interval along a maximal flow, the orbit-saturation step, and the exact use of Proposition 2.6 and Theorem 4.1 of the source.

References

- [1] V. Fischer, M. Ruzhansky, and C. A. Taranto, *Subelliptic Gevrey spaces*, arXiv:1805.08667, especially Introduction Question 3, Proposition 2.6, and Theorem 4.1.
- [2] H. J. Sussmann, *Orbits of families of vector fields and integrability of distributions*, Trans. Amer. Math. Soc. 180 (1973), 171–188.